SMPTE Public Committee Draft

Catena — REST Connection Type

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Foreword

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SMPTE Engineering Documents are drafted in accordance with the rules given in its Standards Operations Manual. This SMPTE Engineering Document was prepared by Technology Committee TC34CS – Media Systems, Control and Services.

Introduction

This clause is entirely informative and does not form an integral part of this Engineering Document.

This document specifies how to transport Catena messages, as defined in SMPTE ST 2138-10, using REST.

1 Scope

This document specifies the use of a REST connection manager with Catena.

2 Conformance

Normative text is text that describes elements of the design that are indispensable or contains the conformance language keywords: "shall", "should", or "may". Informative text is text that is potentially helpful to the user, but not indispensable, and can be removed, changed, or added editorially without affecting interoperability. Informative text does not contain any conformance keywords.

All text in this document is, by default, normative, except: the Introduction, any clause explicitly labeled as "Informative" or individual paragraphs that start with "Note:"

The keywords "shall" and "shall not" indicate requirements strictly to be followed in order to conform to the document and from which no deviation is permitted.

The keywords, "should" and "should not" indicate that, among several possibilities, one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required; or that (in the negative form) a certain possibility or course of action is deprecated but not prohibited.

The keywords "may" and "need not" indicate courses of action permissible within the limits of the document.

The keyword "reserved" indicates a provision that is not defined at this time, shall not be used, and may be defined in the future. The keyword "forbidden" indicates "reserved" and in addition indicates that the provision will never be defined in the future.

A conformant implementation according to this document is one that includes all mandatory provisions ("shall") and, if implemented, all recommended provisions ("should") as described. A conformant implementation need not implement optional provisions ("may") and need not implement them as described.

Unless otherwise specified, the order of precedence of the types of normative information in this document shall be as follows: Normative prose shall be the authoritative definition; tables shall be next; then formal languages; then figures; and then any other language forms.

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3 Normative References

The following standard contains provisions that, through reference in this text, constitute provisions of this standard. Dated references require that the specific edition cited shall be used as the reference. Undated citations refer to the edition of the referenced document (including any amendments) current at the date of publication of this document. All standards are subject to revision, and users of this engineering document are encouraged to investigate the possibility of applying the most recent edition of any undated reference.

SMPTE ST 2138-10:202x, *Catena — Model*

SMPTE ST 2138-11:202x, *Catena — gRPC Connection*

SMPTE ST 2138-19:202x, *Catena — Protocol Objects*

SMPTE ST 2138-50:202x, *Catena — Authenticity - Integrity - Access Control - Confidentiality and Availability*

4 Precedence

In the event of a conflict between the schema located at additional element at https://github.com/SMPTE/st2138-a.git (in the /interface/schemata/device.yaml file) and other information in this document, the schema shall take precedence. See Annex A for additional information.

5 Terms and Definitions

For the purposes of this document, the following terms and definitions apply:

5.1
Representational State Transfer
REST

API that uses HTTP requests to retrieve data from web services that use the REST architectural style

6 The Catena Service / Device Architecture

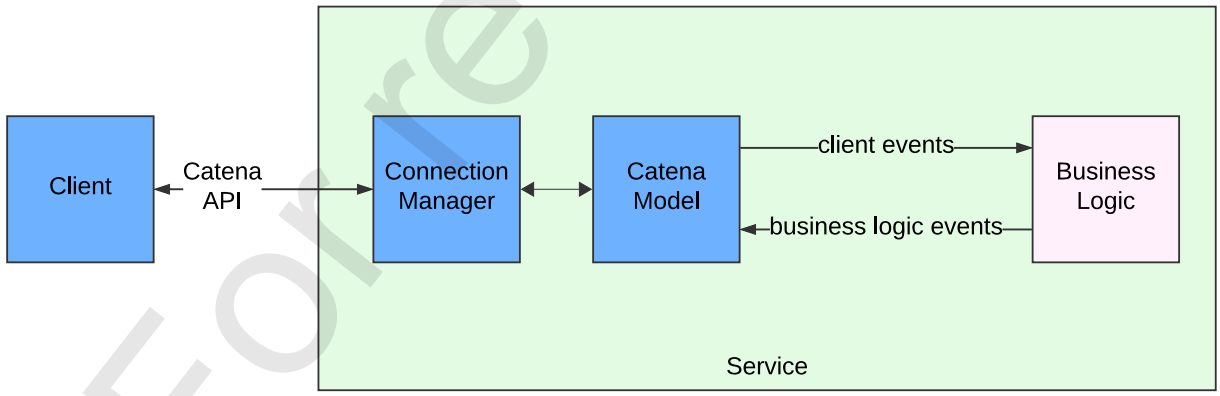

Figure 1 — Catena Device Architecture

Figure 1 provides a high-level view of a Catena compatible device's architecture and the Connection Manager's place within it.

The objective is to isolate the model from both the device's Business Logic and the Connection Manager, which could be implemented using REST (defined in this document), or gRPC (defined in SMPTE ST 2138-11). The Catena Model as defined in SMPTE ST 2138-10 defines the schemata of the objects that travel across the Connection.

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7 REST Connection Type

7.1 Resource Based API

The Catena Model is represented as these resources:

- Discovery – how a client can discover which slots are occupied by a device model and learn the details of each
- Commands – the stateless commands provided by a device model
- Assets – objects that are better not modeled as either parameters or commands – examples would include icon files, log files, and possibly audio or video streams.
- Parameters – the stateful controls, configurations and monitoring information provided by the device model
- Subscriptions – how clients can specify in which parts of the device model they are (and are not) interested. A means of throttling network traffic.
- Updates – notifications of changes to the device's state
- Languages – discovery and field maintenance of a device's language support

The remainder of this document defines the HTTP verbs, together with the structure of the paths and parameters that allow them to interact with each of these resources.

When running in production with authorization enabled, each request shall include an Access Token as specified in SMPTE ST 2138-50 Catena — Authenticity – Integrity - Access Control - Confidentiality and Availability, 202x.

NOTE 1 Objects communicated between client and server are the same as those defined in the gRPC API to which readers are referred for the UML definitions in SMPTE ST 2138-11, unless otherwise stated here.

NOTE 2 A Swagger / OpenAPI rendering of each endpoint specified below in Clause 7.5 is available at https://petstore.swagger.io/?url=https%3A%2F%2Fsmpte.github.io%2Fst2138-a%2Fdocs%2Fopenapi.yaml. Caveat: The OpenAPI is planned to move to a SMPTE administered GitHub repository.

7.2 Default Settings for all Endpoints

Unless superseded in the definition the following settings are common across all endpoints:

- API Title: Catena API
- API Version: 1.0
- Security: The bearer token shall contain at least one of these access scopes: `st2138:mon`, `st2138:op`, `st2138:cfg`, `st2138:adm`, as governed by OAuth2's client credentials flow. See SMPTE ST 2138-50 Catena — Authenticity - Integrity - Access Control - Confidentiality and Availability, 202x for details.

  NOTE All of these scopes are read-only.

- URL preamble: URLs shall use the form: `http(s)://example.com/st2138-api/v1` to allow for updates and improvements to the API, and to avoid name clashes should a service with to offer multiple different APIs.

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7.3 Frequently Used Failure Response Codes

Unless superseded in the path definition, each path shall signal a failed request with the codes defined in Table 1.

Table 1 — Request Failure Response Codes

Code	Meaning
404	Not Found. The requested resource was not found.
401	Unauthorized, the access token is either missing or invalid.
403	Forbidden. Insufficient permission for access.
500	Internal Server Error. An error occurred while processing the request.
503	Service Unavailable. The service is temporarily unavailable.

7.4 Health

This endpoint enables a client to test the continued availability of a service. If no response is obtained, the service is not healthy.

Endpoint 7.4-1 — GET /health

Summary	Health check. If no response provided, the service is not healthy.		
Path	`/health`		
Verb	GET		
Security	Default		
Parameters	None		
Successful Response	**Code**	**Payload**	**Example**
	200	`text/plain`	`OK`
Failure Responses	Default		

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7.5 Discovery

These endpoints enable a client to discover which slots are occupied by device models, and to request the device models themselves.

Endpoint 7.5-1 — GET /devices

Summary	Gets a list of slot numbers that are occupied by a device		
Path	`/devices`		
Verb	GET		
Security	Default		
Parameters	None		
Successful Response	**Code**	**Payload**	**Example**
	200	`application/json`	`"slots": [2,3,5,7,11]`
Failure Responses	Default		

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Endpoint 7.5-2 — GET /{slot}

Summary	Requests a device model to be delivered in a single response. NOTE Device models can be very large, and so clients should consider using the streaming endpoint instead.			
Path	`/{slot}`			
Verb	GET			
Security	Default			
Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
Detail-Level	header	no	Amount of information to include in the response.	SMPTE ST 2138-10 Figure 5, DetailLevel

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json`	

```
"device": {
  "slot": 1,
  "multiSetEnabled": true,
  "subscriptions": true,
  "accessScopes": [
    "st2138:mon",
    "st2138:op",
    "st2138:cfg",
    "st2138:adm"
  ],
  "defaultScope": "st2138:op"
}
```

Failure Responses	Default

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Endpoint 7.5-3 — GET /{slot}/stream

Summary	Requests a device model to be delivered as a stream of Server Sent Events, each a DeviceComponent serialized as JSON and stringified. See SMPTE ST 2138-19 Clause 6.3.
Path	`/{slot}/stream`
Verb	GET
Security	Default

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
Detail-Level	header	no		SMPTE ST 2138-10 Figure 5, DetailLevel

Successful Response	**Code**	**Payload**	**Example**

	200	`text/event-stream`
		
		schema: `string`
			`Param: <stringified ComponentParam>`
			`Menu: <stringified ComponentMenu>`
			`etc.: <more DeviceComponents>`
			`done`

Failure Responses	Default		

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7.6 Commands

Endpoint 7.6-1 — POST /{slot}/command/{fqoid}

Summary	Execute one of the commands advertised by the device model within its Commands object		
Path	`/{slot}/command/{fqoid}`		
Verb	POST		
Request Body	**Description**	**Required**	**Content**
	Optional value to pass to the command, e.g., play speed	No	`application/json` schema: SMPTE ST 2138-19 Clause 6.4
Security	The command's scope (which could be any of the four with write access) shall be present in the access token.		

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	yes	JSON pointer to the asset relative to the device's static root	SMPTE ST 2138-10 Definitions
respond	query	no	Flag to suppress the response when set false	`"true"` or `"false"` (default)

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json` schema SMPTE ST 2138-19 Clause 6.4	`{"string_value": "Hello, World!"}`
Failure Responses	Default		

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Endpoint 7.6-2 — POST /{slot}/command/{fqoid}/stream

Summary	Execute one of the commands advertised by the device model within its Commands object. The command can generate a stream of responses. For example, if the command were “Download Photo Album”, the image files would be sent one at a time until complete.		
Path	`/{slot}/command/{fqoid}/stream`		
Verb	POST		
Request Body	Description	Required	Content
	Optional value to pass to the command, e.g., a folder name	No	application/json schema: `device.yaml#/$defs/value`
Security	The command’s scope (which could be any of the four with write access) shall be present in the access token.		

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	yes	JSON pointer to the asset relative to the device’s static root	SMPTE ST 2138-10 Definitions
respond	query	no	Flag to suppress the response when set false	`"true"` or `"false"` (default)

Successful Response	**Code**	**Payload**	**Example**

	200	`text/event`
		stringified `CommandResponse` objects
		
		schema SMPTE ST 2138-19 Figure 4
			`value: <stringified CommandResponse>`
			`value: <stringified CommandResponse>`
			`done`

Failure Responses	Default		

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7.7 Assets

Assets map to the ExternalObjectRequest RPC within SMPTE ST 2138-11.

Endpoint 7.7-1 — GET /{slot}/asset/{fqoid}

Summary	Get an asset in a single response			
Path	`/{slot}/asset/{fqoid}`			
Verb	GET			
Security	Default			
Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	No	JSON pointer to the asset relative to the device's static root	SMPTE ST 2138-10 Definitions

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json` schema SMPTE ST 2138-10 Figure 5	`{"url": https://example.com/data}`
Failure Responses	Default		

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Endpoint 7.7-2 — POST /{slot}/asset/{fqoid}

Summary	Load an asset onto the service or device
Path	`/{slot}/asset/{fqoid}`
Verb	POST
Security	One of these scopes shall be present in the access token and it shall match the asset's scope: `st2138:adm:w`, `st2138:cfg:w`, `st2138:op:w`. NOTE The monitor scope is not valid for this.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	No	JSON pointer to the asset relative to the device's static root	SMPTE ST 2138-10 Definitions

Request Body	Description	Required	Content
	The external object to load	Yes	`application/json` schema: DataPayload in SMPTE ST 2138-19 Clause 6.6

Successful Response	Code	Payload	Example
	204	Empty	
Failure Responses	Default		

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Endpoint 7.7-3 — PUT /{slot}/asset/{fqoid}

Summary	Overwrite an Asset
Path	`/{slot}/asset/{fqoid}`
Verb	PUT
Security	One of these scopes shall be present in the access token and it shall match the asset's scope: `st2138:adm:w`, st2138:cfg:w, `st2138:op:w`. NOTE That the monitor scope is not valid for this.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	No	JSON pointer to the asset relative to the device's static root	SMPTE ST 2138-10 Definitions

Request Body	**Description**	**Required**	**Content**
	The external object to load	Yes	`application/json` schema: DataPayload in SMPTE ST 2138-19 Figure 6

Successful Response	**Code**	**Payload**	**Example**
	204	Empty	
Failure Responses	Default		

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Endpoint 7.7-4 — GET /{slot}/asset/{fqoid}/stream

Summary	Retrieve one or more assets as a Server Sent Events stream
Path	`/{slot}/asset/{fqoid}/stream`
Verb	GET
Security	One of these scopes shall be present in the access token and it shall match the asset's scope: `st2138:adm:w`, `st2138:cfg:w`, `st2138:op:w`. NOTE The monitor scope is not valid for this.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	No	JSON pointer to the asset, relative to the device's static root	SMPTE ST 2138-10 Definitions

	Code	Payload	Example

Successful Response	200
	
	Schema: `device.yaml#/$defs/data_payload`
		`text/event-stream`
		
		schema string
			`/tom.png: <stringified DataPayload>`
			`/jerry.png: <stringified DataPayload>`
			`done`

Failure Responses	Default		

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7.8 Parameters

The endpoints in this clause allow a client to get information about a device's parameters, and to alter their values.

Endpoint 7.8-1 — GET /{slot}/basic-param/{fqoid}/stream

Summary	Retrieve summary information about a parameter and, optionally, its children as a Server Sent Events stream. If recursive is false, the stream will terminate after the first object has been sent.
Path	`/{slot}/asset/{fqoid}/stream`
Verb	GET
Security	Default. The requested parameter's scope shall be included in the access token's scopes claim.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
recursive	query	no	recursively get info for all children	string `enum ["true", "false"]` default `"false"`
fqoid	path	yes	JSON pointer to the required parameter	SMPTE ST 2138-10 Definitions

Successful Response	Code	Payload	Example

	200	`text/event-stream`
		schema string
		schema of stringified objects: SMPTE ST 2138-19 Figure 7
			`<stringified {"info": BasicParamInfo object, "array_length": 1>`
			`<stringified {"info": BasicParamInfo object, "array_length": 10>`
			`done`

Failure Responses	Default		

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Endpoint 7.8-2 — GET /{slot}/basic-param/{fqoid}

Summary	Retrieve summary information about a parameter and, optionally, its children as a single object.
Path	`/{slot}/asset/{fqoid}`
Verb	GET
Security	Default. The requested parameter's scope shall be included in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	yes	JSON pointer to the required parameter	SMPTE ST 2138-10 Definitions

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json` schema: SMPTE ST 2138-19 Figure 8	

```
{
  "info": {
    "oid": "text_box",
    "name": {
      "display_strings": {
        "en": "Text Box",
        "fr": "Boîte de Texte",
        "es": "Caja de Texto"
      }
    },
    "type": "STRING",
    "template_oid": ""
  }
}
```

Failure Responses	Default

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Endpoint 7.8-3 — GET /{slot}/value/{fqoid}

Summary	Get the requested parameter's value
Path	`/{slot}/value/{fqoid}`
Verb	GET
Security	Default. The requested parameter's scope shall be included in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	yes	JSON pointer to the required parameter	SMPTE ST 2138-10 Definitions

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json` schema: SMPTE ST 2138-10 Figure 8	`{ "string_value": "Hello, World!" }`
Failure Responses	Default		

Endpoint 7.8-4 — PUT /{slot}/value/{fqoid}

Summary	Update the indicated parameter's value
Path	`/{slot}/value/{fqoid}`
Verb	PUT
Security	The requested parameter's scope with write access shall be included in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	yes	JSON pointer to the required parameter	SMPTE ST 2138-10 Definitions

Request Body	**Description**	**Required**	**Content**
	the parameter's new value	yes	`application/json` schema `Value` object in SMPTE ST 2138-10 Figure 8

Successful Response	**Code**	**Payload**	**Example**
	204		
Failure Responses	Default		

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Endpoint 7.8-5 — PUT /{slot}/values/{fqoid}

Summary	Set the value of multiple parameters simultaneously. This is an idempotent operation. In the scenario in which one of the parameters cannot be updated, e.g., due to a bounds violation or access failure, none of the parameters shall be updated.
Path	`/{slot}/value/{fqoid}`
Verb	PUT
Security	The scopes of all of the parameters in the request shall be present with write access in the access token's scope claim.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`

Request Body	Description	Required	Content
	An array of `fqoid` – `Value-object` pairs.	yes	`application/json` SMPTE ST 2138-19 Clause 6.12

Successful Response	Code	Payload	Example
	204		
Failure Responses	Default		

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Endpoint 7.8-6 — GET /{slot}/param/{fqoid}

Summary	Get the requested parameter's full descriptor. Compare with basic-param which just returns summary information.
Path	`/{slot}/param/{fqoid}`
Verb	GET
Security	Default. The requested parameter's scope shall be included in the access token's scopes claim.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
fqoid	path	yes	JSON pointer to the required parameter	SMPTE ST 2138-10 Definitions

Successful Response	Code	Payload	Example
	200	`application/json` schema: `device.yaml#/$defs/component_param`	

```
{
  "oid": "/text_box",
  "param": {
    "name": {
      "display_strings": {
        "en": "Text Box",
        "es": "Caja de Texto",
        "fr": "Boîte de Texte"
      },
      "type": "STRING",
      "value": {
        "string_value": "Hello!"
      }
    }
  }
}
```

Failure Responses	Default

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7.9 Subscriptions

The endpoints in this clause enable a client to choose parameters for which to receive updates. This is a way to reduce network traffic.

Endpoint 7.9-1 — GET /{slot}/subscriptions

Summary	Get the list of currently subscribed parameters.
Path	`/{slot}/subscriptions`
Verb	GET
Security	Default. The requested parameter's scope shall be included in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json` schema: array of `component_param`	

```
[
  {
    "fqoid": "text_box",
    "param":
    "fqoid_aliases": [],
    "client_hints": {},
    "commands": {},
    "params": {},
    "name": {
      "display_strings": {
        "fr": "Boîte de Texte",
        "en": "Text Box",
        "es": "Caja de Texto"
      }
    },
    "type": "STRING",
    "read_only": false,
    "value": {
      "string_value":
        "Hello, World!"
    },
  }
  }
]
```

Failure Responses	Default

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Endpoint 7.9-2 — PUT /{slot}/subscriptions

Summary	Remove and add oids to the current set of subscribed oids			
Path	`/{slot}/subscriptions`			
Verb	PUT			
Security	Default			
Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
Request Body	Description	Required	Content	
	An array pair of oids to add and oids to remove from the subscription	yes	`application/json`	
Successful Response	**Code**	**Payload**	**Example**	
	204			
Failure Responses	Default			

7.10 Updates

This endpoint enables a client to subscribe to a stream of updates about changes in the device's state.

Endpoint 7.10-1 — GET /connect

Summary	Connects a client to a service for updates on state changes			
Path	`/{slot}/connect`			
Verb	GET			
Security	Default.			
Parameters	**Location**	**Required**	**Description**	**Schema**
Detail-Level	header	no	The level of detail to provide in the updates	`device.yaml#/$refs/detail_level`
Successful Response	**Code**	**Payload**	**Example**	

	200	`test/event-stream`	`value: <stringified Value>`
			`exception: <stringified Exception>`
			`done`	

Failure Responses	Default			

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7.11 Languages

The endpoints in this clause enable a client to get information about the languages supported by the device, and to add / remove additional language packs in the field.

Endpoint 7.11-1 — GET /{slot}/languages

Summary	Connects a client to a service for updates on state changes
Path	`/{slot}/connect`
Verb	GET
Security	Default. The requested parameter's scope shall be included in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`

Successful Response	**Code**	**Payload**	**Example**
	200	`application/json` schema: SMPTE ST 2138-19 Clause 6.10	`{ "languages": [ "en", "fr", "es", "de" ] }`
Failure Responses	Default		

For review only

Endpoint 7.11-2 — GET /{slot}/language-pack/{language-code}

Summary	Get the language pack specified by the language code
Path	`/{slot}/connect`
Verb	GET
Security	Default.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
language-code	path	yes	standard language identifying code	string

Successful Response	Code	Payload	Example

	200	`application/json`
		schema:
		`device.yaml#/$defs/language_pack`
			`{ "name": "Spanish", "words": { "greeting": "Hola", "parting": "Adiós" } }`

Failure Responses	Default		

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Endpoint 7.11-3 — PUT /{slot}/language-pack/{language-code}

Summary	Update a language pack in the device model. Only packs that are field upgrades can be updated. Attempts to update a pack that was shipped with the device shall fail with 403 Forbidden.
Path	`/{slot}/language-pack/{language-code}`
Verb	PUT
Security	`st2138:adm:w` shall be present in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
language-code	path	yes	standard language identifying code	string

Request Body	**Description**	**Required**	**Content**
	A language pack	yes	`application/json` schema: `device.yaml#/$refs/language_pack`

	Code	**Payload**	**Example**
Successful Response	204		
Failure Responses	Default		

Endpoint 7.11-4 — POST /{slot}/language-pack/{language-code}

Summary	Add a new language pack to the device model. Attempts to add a factory-set language pack that was shipped with the device shall fail with 403 Forbidden.
Path	`/{slot}/language-pack/{language-code}`
Verb	POST
Security	`st2138:adm:w` shall be present in the access token's scopes claim.

Parameters	**Location**	**Required**	**Description**	**Schema**
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
language-code	path	yes	standard language identifying code	string

Request Body	**Description**	**Required**	**Content**
	A language pack	yes	`application/json` schema: `device.yaml#/$refs/language_pack`

	Code	**Payload**	**Example**
Successful Response	204		
Failure Responses	Default		

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Endpoint 7.11-5 — DELETE /{slot}/language-pack/{language-code}

Summary	Remove a field-installed language pack. Attempts to remove factory-set language packs shall fail with 403 Forbidden.
Path	`/{slot}/language-pack/{language-code}`
Verb	DELETE
Security	`st2138:adm:w` shall be present in the access token's scopes claim.

Parameters	Location	Required	Description	Schema
slot	path	yes	Locally unique identifier	`device.yaml#/$defs/slot`
language-code	path	yes	standard language identifying code	string

	Code	Payload	Example
Successful Response	204		
Failure Responses	Default		

Endpoint 7.11-6 — GET /access

Summary	Refresh the access token with the one attached in the request's metadata.
Path	`/access`
Verb	GET
Security	Sub field shall be the same as the current or expiring token that is currently keeping active connections alive.

Parameters	Location	Required	Description	Schema
none				

	Code	Payload	Example
Successful Response	200	`application/json` `#/$defs/connection_status`	`{ "connected": true, "exp": 171641600 }`
Failure Responses	Default		

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Endpoint 7.11-7 — POST /access

Summary	Revoke access for subjects named in the payload.
Path	`/access`
Verb	POST
Security	`st2138:adm:w` shall be present in the access token's scopes claim.

Parameters	Location	Required	Description	Schema
none				

Request Body	Description	Required	Content
	Revoke Access Payload	yes	application/json schema: `#/$defs/revoke_access_payload`

Successful Response	Code	Payload	Example
	200	`application/json` schema: `#/$defs/revocation_response`	`{ "remaining_subjects": [ "user123", "user456" ], "revoked_subjects": [ "user789" ] }`
Failure Responses	Default		

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Annex A
(Informative)

Additional Elements

A.1 Repository

A set of schemas represented by the UML diagrams in this document is provided as a companion element to this document. These can be found in the public repository at the following URL:

https://github.com/SMPTE/st2138-a.git

The contents of st2138-a.git are normative.

The software version corresponding to this document revision is tagged as v1.0.0-pcd.

A.2 Root Directory

Throughout this document, the value of the symbol $(ROOT) is the directory into which the test materials are cloned from the repository.

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